Faculty of Computing

Universiti Teknologi Malaysia

Data Structure & Algorithm

Lab Exercise 01

Instruction

- **Write down** all answers for each question in Part 1-3. Upload the answers (scanned photo) in E-learning Forum by replying to the main thread with your name, matric no and perform submission of the source code (.cpp) for Part 4.

Part 1:

1. Given the following program, show the complete recursive process for the answer find (1,5).

```
1   int find(int a,int b) {
2   if (a==0)
3   {
4   return (b+1);
5   }
6   if(b==0)
7   {
8   return(find(a-1,1));
9   }
10  else
11  {
12  return (find(a-1,find(a,b-1)));
13  }
14 }
```

Write down the trace using:

```
int find(1,5) {
if (___==0)
{
return (___+1);
}
if(___==0)
{
return(find(___-1,1));
}
else
{
return (find(___-1,find(___,___-1)));
}
}
```

return 7

(find(1-1), find(1,4))) + 1

(find(1-1), find(1,3))) + 1

(find(1-1), find(1,2))) + 1

(find(1-1), find(1,1))) + 1

(find(1-1), find(1,0))) + 1

2+5 = 7

(find(0,1) ⟶ a=0, b=1

b+1 = 1+1

= 2

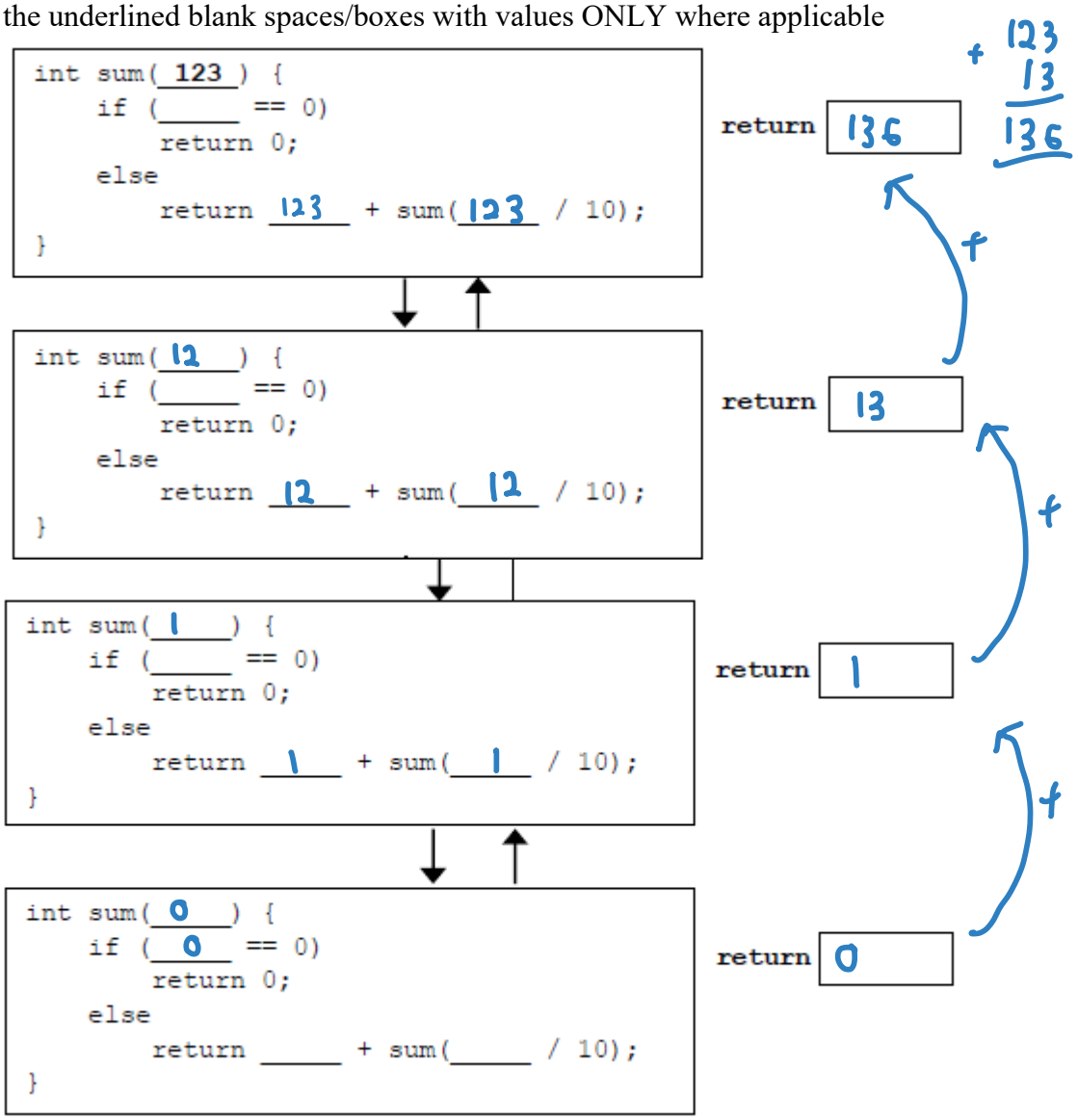
return 2

Part 2:

1. Given the following program, Trace the recursive calls of the sum function in the above program. Write your answer by completing the tracing template below when n = 123.

```
1   int sum(int n) {
2   if (n == 0)
3   return 0;
4   else
5   return n + sum(n / 10);
6   }
7   int main() {
8   cout << sum(123);
9   return 0;
10  }
```

Fill in the underlined blank spaces/boxes with values ONLY where applicable



Part 3:

1. Given the following program:

```
1   #include <iostream>
2   using namespace std;
3
4   void reverseString(string str, int index) {
5       if (index < 0)
6           return;
7       cout << index << ":" << str[index] << "\n";
8       reverseString(str, index - 1);
9   }
10
11  int main() {
12      string text = "ALGORITHM";
13      reverseString(text, text.length() - 1);
14      return 0;
15  }
```

a. Trace the recursive algorithm and write the output.

b. What is the base/terminal case of the `reverseString` function

c. What is the general case of the `reverseString` function

8 : M
7 : H
6 : T
5 : I
4 : R
3 : O
2 : G
1 : L
0 : A

b) if(index < 0)

c) reverseString(str, index - 1)

0 1 2 3 4 5 6 7 8
A L G O R I T H M